

Ylya Martchenko

IT LEAD ENGINEER | CLOUD, IOT & SOFTWARE ARCHITECTURE

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French Citizen (EU)

Available from July 13th, 2026

LANGUAGES

French

C2 • Native

English

C1 • Fluent

Russian

C1 • Bilingual

SKILLS

Full Stack Web & Mobile

Next.js

Angular

React Native

Liquid Templates

JavaScript / ES6

Tailwind CSS

WebSocket / Server-Sent

Backend & Integration

Spring Boot

Microservices

RabbitMQ / Kafka

REST APIs / GraphQL

PostgreSQL / pgvector

MySQL / Redis Cache

Hexagonal Architectures

Clean Architecture / DDD

Cloud & DevOps

CI/CD

Kubernetes (K8s)

Docker / Containers

GitHub Actions / GitLab CI

Jenkins / Ansible / Terraform

Linux / Bash Scripting

Staging / Production

IoT & OT Systems

LIN-bus

CAN-bus

FreeRTOS

ESP32

SPI / I2C / UART

BLE

Modbus TCP/RTU

MQTT / CoAP

Metrology (Trescal)

Languages & Scripting

TS

Java

C++

Python

C

Rust

C# / .NET

SQL / PL/pgSQL

Security, SRE & Observability

OAuth2 / OpenID / JWT

Zero Trust / IAM

NIS2 / GDPR Compliance

WASM Sandbox

Prometheus / Grafana

OpenTelemetry / SRE

AI / RAG Engineering

Git / Postman / Insomnia

Explore live interactive demos & code repositories

hy13dev.com

IT LEAD PROFILE

Computer Science Engineer graduated from Polytech Nancy (IA2R). My background combines solid full-stack engineering expertise (Spring Boot microservices, React/Angular, .NET Core) and secure cloud infrastructure (Kubernetes, CI/CD pipelines) with critical embedded IoT/OT systems (LIN/CAN automotive gateways, real-time physical buses, FreeRTOS). Versatile and adaptable, I bridge business requirements with highly available, resilient, and secure distributed architectures.

PROFESSIONAL EXPERIENCE

Full Stack Developer (Apprenticeship)

Equasens • Villers-lès-Nancy, France

2023/09 — Present

HDS Environment & Compliance:

Developing critical software solutions within an HDS (Health Data Hosting) certified ecosystem, ensuring sensitive data protection and strict compliance with GDPR standards.

Microservices & Cloud Architecture:

Designing and maintaining highly resilient distributed microservices using Spring Boot, orchestrated via Kubernetes (K8s/Docker) and securing asynchronous flows via RabbitMQ.

DevOps & Critical Tasks:

Full automation of CI/CD pipelines under Jenkins and developing automated background CRON jobs for processing and reconciling heavy data under PostgreSQL.

Tech Lead / Full Stack Developer

Domaine Schumacher-Knepper • Wintrange, Luxembourg

2025/06 — 2025/09

Security & Alcohol Compliance:

Designed a custom interception gateway running in the DOM event-capturing phase (useCapture = true) to halt checkout actions, ensuring 100% legal compliance in Luxembourg.

Performance & Zero CLS (Lighthouse 100):

Comprehensive optimization of the static rendering architecture (Liquid/CSS), achieving a 100/100 Lighthouse Performance score and zero Cumulative Layout Shift (CLS) on mobile devices.

IntersectionObserver & Web Component:

Developed a historical countdown component optimized for mobile CPU cycles using IntersectionObserver and synchronized at a constant 60 FPS using requestAnimationFrame.

Compliant with Luxembourgish Alcohol Legislation

Multi-language website: FR | DE | LU | EN

ARCHITECTURE & R&D PROJECTS

TesLimitless — Active LIN/CAN IoT Gateway

Lead Embedded & Mobile Systems Engineer

C++FreeRTOSESP32LIN-busCAN-busReact Native (BLE)

A physical Man-In-The-Middle gateway on the LIN automotive bus of a Tesla Model 3 Highland, intercepting capacitive steering wheel button commands to inject new macros on the CAN-bus via the diagnostic port. Coupled via BLE to a custom React Native mobile application displaying real-time vehicle telemetry in HUD mode.

S Situation:

Real-time processing obligation on the LIN-bus within a strict deterministic delay of 1.5ms, under penalty of haptic communication errors.

A Action:

Isolation of the safety-critical LIN capture task on Core 1 of the ESP32 using FreeRTOS C++, offloading CAN calculations and BLE telemetry to Core 2.

R Result:

Latency of sub-1.5ms deterministic execution, total stability (0 frame errors), and fluid telemetry.

YlyaBot — RAG AI Agent & Observability Dashboard

Lead Systems Architect & AI Engineer

Next.js 16React 19Supabase RPCpgvectorRedis (Upstash)JWT

Recruitment agent based on decoupled RAG in a Git submodule. Semantic search by cosine distance in a PostgreSQL database, coupled with an analytical observability portal.

S Situation:

Avoid 429 rate limit errors from free LLM quotas during simultaneous user access sessions.

A Action:

Conception of a fallback infrastructure with 5 priority levels, paired with a quota exclusion cookie at midnight and a Redis cache. Native PostgreSQL vector computation via PL/pgSQL RPC under 10ms.

R Result:

100% availability, first token under 150ms, and analytical dashboard secured under JWT.

EDUCATION

Engineer in IA2R (Informatics, Automatic, Robotics, Networks)

Polytech Nancy • Vandœuvre-lès-Nancy, France

2023 — 2026

Diploma of Higher Education in Engineering Science

F.R.U. Mathematics, computer science, mechanics • Metz, France

2022 — 2023

BTEC Higher National Diploma in Physics Measures

U.I.T. Department of Measures • Metz, France

2021 — 2022

PeiP (Polytech engineering schools course)

Polytech Nancy • Vandœuvre-lès-Nancy, France

2019 — 2021